20261006 Quiz

for (MATH230) Probability & Statistics

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Exercise 4.34. Let X be a random variable with the following probability distribution:

x	-2	3	5
$f(x)$	0.3	0.3	0.4

Find the standard deviation of X.

Solution. By Theorem 4.2,

$$\mathbb{E}[X] = (-2)(0.3) + (3)(0.3) + (5)(0.4) = \frac{23}{10},$$
$$\mathbb{E}[X^2] = (-2)^2(0.3) + 3^2(0.3) + 5^2(0.4) = \frac{139}{10}.$$

Hence,

$$\mathrm{Var}(X) = \frac{139}{10} - \left(\frac{23}{10}\right)^2 = \frac{861}{100},$$

so

$$\boxed{\sigma_X = \frac{\sqrt{861}}{10} \approx 2.93}.$$

□

Exercise 4.35. The random variable X, representing the number of errors per 100 lines of software code, has the following probability distribution:

x	2	3	4	5	6
$f(x)$	0.01	0.25	0.4	0.3	0.04

Using Theorem 4.2 on page 141, find the variance of X.

Solution. Theorem 4.2 gives

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \left(\mathbb{E}[X]\right)^2.$$

Here,

$$\mathbb{E}[X] = (2)(0.01) + (3)(0.25) + (4)(0.4) + (5)(0.3) + (6)(0.04) = 4.11,$$
$$\mathbb{E}[X^2] = 2^2(0.01) + 3^2(0.25) + 4^2(0.4) + 5^2(0.3) + 6^2(0.04) = 17.63.$$

Therefore,

$$\boxed{\mathrm{Var}(X) = 17.63 - (4.11)^2 = 0.7379 \approx 0.74}.$$

□

Exercise 4.36. Suppose that the probabilities are 0.4, 0.3, 0.2, and 0.1, respectively, that 0, 1, 2, or 3 power failures will strike a certain subdivision in any given year. Find the mean and variance of the random variable X representing the number of power failures striking this subdivision.

Solution. The distribution is

$$p_X(0) = 0.4, \quad p_X(1) = 0.3, \quad p_X(2) = 0.2, \quad p_X(3) = 0.1.$$

Thus,

$$\mathbb{E}[X] = (1)(0.3) + (2)(0.2) + (3)(0.1) = 1,$$
$$\mathbb{E}[X^2] = (1)^2(0.3) + (2)^2(0.2) + (3)^2(0.1) = 2.$$

By Theorem 4.2,

$$\mathrm{Var}(X) = 2 - 1^2 = 1.$$

Hence,

$$\boxed{\mu_X = 1, \qquad \mathrm{Var}(X) = 1}.$$

□

Exercise 4.37. A dealer's profit on a new automobile, measured in units of \$5000, is a random variable X with density

$$f_X(x) = \begin{cases} 2(1-x), & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the variance of X.

Solution. The given density is

$$f(x) = \begin{cases} 2(1-x), & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Direct integration gives

$$\mathbb{E}[X] = \int_0^1 2x(1-x)\ \mathrm{d}x = \frac{1}{3},$$
$$\mathbb{E}[X^2] = \int_0^1 2x^2(1-x)\ \mathrm{d}x = \frac{1}{6}.$$

Therefore,

$$\boxed{\mathrm{Var}(X) = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{18}}.$$

Since X is measured in units of \$5000, the variance of the actual profit in dollars is

$$\frac{1}{18}(5000)^2 \text{ dollars}^2.$$

□

Exercise 4.40. Let X have density

$$f_X(x) = \begin{cases} \dfrac{2(x+2)}{5}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find $\mathrm{Var}(g(X))$ for $g(X) = 3X + 4$.

Solution. The given density is

$$f(x) = \begin{cases} \dfrac{2(x+2)}{5}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

By Theorem 4.3,

$$\mathrm{Var}(g(X)) = \mathbb{E}[g(X)^2] - \left(\mathbb{E}[g(X)]\right)^2.$$

The required moments are

$$\mathbb{E}[g(X)] = \int_0^1 (3x+4)\frac{2(x+2)}{5}\,\mathrm{d}x = \frac{28}{5},$$
$$\mathbb{E}[g(X)^2] = \int_0^1 (3x+4)^2\frac{2(x+2)}{5}\,\mathrm{d}x = \frac{321}{10}.$$

Therefore,

$$\boxed{\mathrm{Var}(g(X)) = \frac{321}{10} - \left(\frac{28}{5}\right)^2 = \frac{37}{50} = 0.74}.$$

□

Exercise 4.41. Let X have distribution

x	-3	6	9
$p_X(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

Find the standard deviation of $h(X) = (3X+1)^2$.

Solution. The distribution of X is

x	-3	6	9
$p_X(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

and

$$h(-3) = 64, \qquad h(6) = 361, \qquad h(9) = 784.$$

By Theorem 4.3,

$$\mathbb{E}[h(X)] = 64\left(\frac{1}{6}\right) + 361\left(\frac{1}{2}\right) + 784\left(\frac{1}{3}\right) = \frac{905}{2},$$
$$\mathbb{E}[h(X)^2] = 64^2\left(\frac{1}{6}\right) + 361^2\left(\frac{1}{2}\right) + 784^2\left(\frac{1}{3}\right) = \frac{541457}{2}.$$

Thus,

$$\mathrm{Var}(h(X)) = \frac{541457}{2} - \left(\frac{905}{2}\right)^2 = \frac{263889}{4},$$

and therefore

$$\boxed{\sigma_{h(X)} = \frac{\sqrt{263889}}{2} \approx 256.85}.$$

□

Exercise 4.43. The length of time, in minutes, for an airplane to obtain clearance for takeoff at a certain airport is a random variable $Y = 3X - 2$, where X has the density function

$$f(x) = \begin{cases} \frac{1}{4}e^{-x/4}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the mean and variance of the random variable Y.

Solution. Since $Y = g(X) = 3X - 2$, we use the expectation formulas for a function of X directly:

$$\mathbb{E}[Y] = \int_0^\infty (3x - 2)\frac{1}{4}e^{-x/4}\, \mathrm{d}x = 10,$$
$$\mathbb{E}[Y^2] = \int_0^\infty (3x - 2)^2\frac{1}{4}e^{-x/4}\, \mathrm{d}x = 244.$$

Therefore,

$$\mathrm{Var}(Y) = 244 - 10^2 = 144.$$

Hence,

$$\boxed{\mu_Y = 10, \qquad \mathrm{Var}(Y) = 144}.$$

□

Exercise 4.44. Four fruits are sampled without replacement from three oranges, two apples, and three bananas. Let X and Y count the oranges and apples selected, respectively. Find $\mathrm{Cov}(X, Y)$.

Solution. Four pieces are selected from 3 oranges, 2 apples, and 3 bananas, with X and Y denoting the numbers of oranges and apples selected. Thus

$$p_{X,Y}(x, y) = \frac{\binom{3}{x}\binom{2}{y}\binom{3}{4-x-y}}{\binom{8}{4}}$$

for the admissible values of x and y. The marginal distributions give

$$p_X(0) = \frac{1}{14}, \quad p_X(1) = \frac{3}{7}, \quad p_X(2) = \frac{3}{7}, \quad p_X(3) = \frac{1}{14},$$
$$p_Y(0) = \frac{3}{14}, \quad p_Y(1) = \frac{4}{7}, \quad p_Y(2) = \frac{3}{14}.$$

Hence,

$$\mathbb{E}[X] = \frac{3}{2}, \qquad \mathbb{E}[Y] = 1.$$

Also,

$$\mathbb{E}[XY] = \sum_x \sum_y xy\, p_{X,Y}(x, y)$$
$$= \frac{18 + 18 + 36 + 12 + 6}{70} = \frac{9}{7}.$$

Therefore, by Theorem 4.4,

$$\boxed{\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \frac{9}{7} - \frac{3}{2} = -\frac{3}{14}}.$$

□

Exercise 4.45. Let X count machine malfunctions and Y technician calls. Their joint pmf is

$p_{X,Y}(x,y)$	$x=1$	$x=2$	$x=3$
$y=1$	0.05	0.05	0.10
$y=3$	0.05	0.10	0.35
$y=5$	0	0.20	0.10

Find $\mathrm{Cov}(X,Y)$.

Solution. The joint distribution is

$p_{X,Y}(x,y)$	x		
y	1	2	3
1	0.05	0.05	0.10
3	0.05	0.10	0.35
5	0.00	0.20	0.10

so the marginal distributions are

$$p_X(1)=0.10,\quad p_X(2)=0.35,\quad p_X(3)=0.55,$$
$$p_Y(1)=0.20,\quad p_Y(3)=0.50,\quad p_Y(5)=0.30.$$

Thus,

$$\mathbb{E}[X]=\frac{49}{20},\qquad \mathbb{E}[Y]=\frac{16}{5},$$

The mixed moment is

$$\begin{aligned}\mathbb{E}[XY] &= \sum_x\sum_y xy\,p_{X,Y}(x,y)\\ &= (1)(1)(0.05)+(2)(1)(0.05)+(3)(1)(0.10)\\ &\quad +(1)(3)(0.05)+(2)(3)(0.10)+(3)(3)(0.35)\\ &\quad +(2)(5)(0.20)+(3)(5)(0.10)\\ &= \frac{157}{20}.\end{aligned}$$

Therefore, by Theorem 4.4,

$$\boxed{\mathrm{Cov}(X,Y)=\frac{157}{20}-\left(\frac{49}{20}\right)\left(\frac{16}{5}\right)=\frac{1}{100}=0.01}.$$

□

Exercise 4.48. Given a random variable X, with standard deviation σ_X, and a random variable $Y=a+bX$, show that if $b<0$, the correlation coefficient $\rho_{X,Y}=-1$, and if $b>0$, $\rho_{X,Y}=1$.

Solution. Assume $\sigma_X>0$, as required for the correlation coefficient to be defined. By Theorem 4.5,

$$\mathbb{E}[Y]=a+b\mathbb{E}[X].$$

We have

$$Y-\mathbb{E}[Y]=b(X-\mathbb{E}[X]).$$

Hence, directly from the definitions,

$$\operatorname{Cov}(X,Y) = b\operatorname{Var}(X) = b\sigma_X^2,$$
$$\sigma_Y = |b|\sigma_X.$$

Therefore,

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X,Y)}{\sigma_X\sigma_Y} = \frac{b}{|b|} = \begin{cases} -1, & b < 0, \\ 1, & b > 0. \end{cases}$$

□

Exercise 4.52. Random variables X and Y follow a joint distribution

$$f(x,y) = \begin{cases} 2, & 0 < x \le y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the correlation coefficient between X and Y.

Solution. Over the support $0 < x \le y < 1$,

$$\mathbb{E}[X] = \int_0^1 \int_0^y 2x \, \mathrm{d}x \, \mathrm{d}y = \frac{1}{3}, \qquad \mathbb{E}[Y] = \int_0^1 \int_0^y 2y \, \mathrm{d}x \, \mathrm{d}y = \frac{2}{3},$$
$$\mathbb{E}[X^2] = \int_0^1 \int_0^y 2x^2 \, \mathrm{d}x \, \mathrm{d}y = \frac{1}{6}, \quad \mathbb{E}[Y^2] = \int_0^1 \int_0^y 2y^2 \, \mathrm{d}x \, \mathrm{d}y = \frac{1}{2},$$
$$\mathbb{E}[XY] = \int_0^1 \int_0^y 2xy \, \mathrm{d}x \, \mathrm{d}y = \frac{1}{4}.$$

Thus,

$$\operatorname{Var}(X) = \operatorname{Var}(Y) = \frac{1}{18}, \qquad \operatorname{Cov}(X,Y) = \frac{1}{4} - \left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = \frac{1}{36}.$$

Therefore,

$$\boxed{\rho_{X,Y} = \frac{1/36}{\sqrt{(1/18)(1/18)}} = \frac{1}{2}}.$$

□

Exercise 4.57. Let X be a random variable with the following probability distribution:

x	-3	6	9
$f(x)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$

Find $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ and then, using these values, evaluate $\mathbb{E}[(2X+1)^2]$.

Solution. Directly from the distribution,

$$\mathbb{E}[X] = (-3)\left(\frac{1}{6}\right) + 6\left(\frac{1}{3}\right) + 9\left(\frac{1}{2}\right) = 6,$$
$$\mathbb{E}[X^2] = 9\left(\frac{1}{6}\right) + 36\left(\frac{1}{3}\right) + 81\left(\frac{1}{2}\right) = 54.$$

Thus,

$$\boxed{\mathbb{E}[X] = 6, \qquad \mathbb{E}[X^2] = 54}.$$

Since $(2X+1)^2 = 4X^2 + 4X + 1$, Theorem 4.6 gives

$$\boxed{\mathbb{E}[(2X+1)^2] = 4\mathbb{E}[X^2] + 4\mathbb{E}[X] + 1 = 241}.$$

□

Exercise 4.58. The total time, measured in units of 100 hours, that a teenager runs their hair dryer over a period of one year is a continuous random variable X that has the density function

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \le x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Use Theorem 4.6 to evaluate the mean of the random variable $Y = 60X^2 + 48X$, where Y is equal to the number of kilowatt hours expended annually.

Solution. First,

$$\mathbb{E}[X] = \int_0^1 x^2 \, \mathrm{d}x + \int_1^2 x(2 - x) \, \mathrm{d}x = 1,$$
$$\mathbb{E}[X^2] = \int_0^1 x^3 \, \mathrm{d}x + \int_1^2 x^2(2 - x) \, \mathrm{d}x = \frac{7}{6}.$$

By Theorem 4.6,

$$\mathbb{E}[Y] = 60\mathbb{E}[X^2] + 48\mathbb{E}[X] = 60\left(\frac{7}{6}\right) + 48 = \boxed{118}.$$

Thus, the mean annual energy expenditure is 118 kilowatt hours.

□

Exercise 4.60. Let X and Y have the following joint probability distribution:

$f(x, y)$	x	
y	2	4
1	0.15	0.10
3	0.25	0.25
5	0.15	0.10

Find

(a) $\mathbb{E}[2X - 3Y]$;
(b) $\mathbb{E}[XY]$.

Solution. The marginal means are

$$\mathbb{E}[X] = (2)(0.55) + (4)(0.45) = \frac{29}{10}, \qquad \mathbb{E}[Y] = (1)(0.25) + (3)(0.50) + (5)(0.25) = 3.$$

(a) By Theorems 4.5 and 4.7,

$$\boxed{\mathbb{E}[2X - 3Y] = 2\mathbb{E}[X] - 3\mathbb{E}[Y] = -\frac{16}{5}}.$$

(b) Using the displayed joint distribution directly,

$$\begin{aligned} \mathbb{E}[XY] &= \sum_x \sum_y xyf(x, y) \\ &= (2)(1)(0.15) + (4)(1)(0.10) + (2)(3)(0.25) + (4)(3)(0.25) \\ &\quad + (2)(5)(0.15) + (4)(5)(0.10) \\ &= \boxed{\frac{87}{10}}. \end{aligned}$$

□

Exercise 4.61. Use Theorem 4.7 to evaluate $\mathbb{E}[2XY^2 - X^2Y]$ for the joint probability distribution shown in Table 3.1 on page 116.

Solution. By Theorem 4.7,

$$\mathbb{E}[2XY^2 - X^2Y] = 2\mathbb{E}[XY^2] - \mathbb{E}[X^2Y].$$

Both terms contain a factor XY, so every cell with $x = 0$ or $y = 0$ contributes 0. From Table 3.1, the only remaining nonzero cell is

$$p_{X,Y}(1,1) = \frac{3}{14}.$$

Hence,

$$\mathbb{E}[XY^2] = \frac{3}{14}, \qquad \mathbb{E}[X^2Y] = \frac{3}{14},$$

and therefore

$$\boxed{\mathbb{E}[2XY^2 - X^2Y] = \frac{3}{14}}.$$

□

Exercise 4.65. Let X represent the number that occurs when a red die is tossed and Y the number that occurs when a green die is tossed. Find

(a) $\mathbb{E}[2X + Y]$;
(b) $\mathbb{E}[X - 2Y]$;
(c) $\mathbb{E}[2XY]$.

Solution. For each die,

$$\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1}{6}\sum_{k=1}^{6} k = \frac{7}{2}.$$

Therefore, by Theorems 4.5 and 4.7,

$$\text{(a)} \quad \mathbb{E}[2X + Y] = 2\mathbb{E}[X] + \mathbb{E}[Y] = \boxed{\frac{21}{2}},$$

$$\text{(b)} \quad \mathbb{E}[X - 2Y] = \mathbb{E}[X] - 2\mathbb{E}[Y] = \boxed{-\frac{7}{2}}.$$

For part (c), the two die outcomes are independent, so their joint pmf is $1/36$ on $\{1, \ldots, 6\}^2$. Thus

$$\mathbb{E}[XY] = \frac{1}{36}\left(\sum_{x=1}^{6} x\right)\left(\sum_{y=1}^{6} y\right) = \frac{49}{4},$$

$$\text{(c)} \quad \mathbb{E}[2XY] = \boxed{\frac{49}{2}}.$$

□

Exercise 4.69. The random variables X and Y represent the number of vehicles that arrive at two separate street corners during a certain 2-minute period in the day. The joint distribution is

$$f(x,y) = \left(\frac{1}{4^{x+y}}\right)\left(\frac{9}{16}\right),$$

for $x = 0, 1, 2, \ldots$ and $y = 0, 1, 2, \ldots$.

(a) Give $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathrm{Var}(X)$, and $\mathrm{Var}(Y)$.
(b) Consider $Z = X + Y$, the sum of the two. Find $\mathbb{E}[Z]$ and $\mathrm{Var}(Z)$.

Solution. The joint pmf factors as

$$f(x,y) = \left(\frac{3}{4}\left(\frac{1}{4}\right)^x\right)\left(\frac{3}{4}\left(\frac{1}{4}\right)^y\right),$$

so X and Y are independent and have the same marginal pmf.
(a) For $|r| < 1$,

$$\sum_{k=0}^{\infty} kr^k = \frac{r}{(1-r)^2}, \qquad \sum_{k=0}^{\infty} k^2 r^k = \frac{r(1+r)}{(1-r)^3}.$$

With $r = 1/4$,

$$\mathbb{E}[X] = \mathbb{E}[Y] = \frac{3}{4}\sum_{k=0}^{\infty} k\left(\frac{1}{4}\right)^k = \frac{1}{3},$$

$$\mathbb{E}[X^2] = \mathbb{E}[Y^2] = \frac{3}{4}\sum_{k=0}^{\infty} k^2\left(\frac{1}{4}\right)^k = \frac{5}{9}.$$

Therefore,

$$\boxed{\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1}{3}, \qquad \mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{4}{9}}.$$

(b) By Theorem 4.7,

$$\mathbb{E}[Z] = \mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] = \frac{2}{3}.$$

For the mixed moment, use the factorized joint pmf directly:

$$\begin{aligned}\mathbb{E}[XY] &= \sum_x \sum_y xy\left(\frac{3}{4}\left(\frac{1}{4}\right)^x\right)\left(\frac{3}{4}\left(\frac{1}{4}\right)^y\right) \\ &= \left(\frac{3}{4}\sum_{x=0}^{\infty} x\left(\frac{1}{4}\right)^x\right)\left(\frac{3}{4}\sum_{y=0}^{\infty} y\left(\frac{1}{4}\right)^y\right) = \frac{1}{9}.\end{aligned}$$

Hence,

$$\mathbb{E}[Z^2] = \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] = \frac{4}{3},$$

$$\mathrm{Var}(Z) = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 = \frac{4}{3} - \left(\frac{2}{3}\right)^2 = \frac{8}{9}.$$

Therefore,

$$\boxed{\mathbb{E}[Z] = \frac{2}{3}, \qquad \mathrm{Var}(Z) = \frac{8}{9}}.$$

□

Exercise 4.70. There are two service lines. The random variables X and Y are the proportions of time that line 1 and line 2 are in use, respectively. The joint probability density function for (X, Y) is given by

$$f(x,y) = \begin{cases} \frac{3}{2}(x^2+y^2), & 0 \le x, y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Determine whether or not X and Y are independent.
(b) For $Z = X + Y$, the sum of the two utilization proportions, find $\mathbb{E}[Z]$ and $\mathbb{E}[XY]$.
(c) Find $\mathrm{Var}(X)$, $\mathrm{Var}(Y)$, and $\mathrm{Cov}(X, Y)$.
(d) Find $\mathrm{Var}(X + Y)$.

Solution. (a) The marginal densities are

$$f_X(x) = \int_0^1 \frac{3}{2}(x^2 + y^2)\ \mathrm{d}y = \frac{1 + 3x^2}{2},$$
$$f_Y(y) = \int_0^1 \frac{3}{2}(x^2 + y^2)\ \mathrm{d}x = \frac{1 + 3y^2}{2},$$

for $0 \le x, y \le 1$. Since $f(x, y) \neq f_X(x) f_Y(y)$ in general,

$$\boxed{X \text{ and } Y \text{ are not independent.}}$$

(b) By symmetry,

$$\mathbb{E}[X] = \mathbb{E}[Y] = \int_0^1 x \frac{1 + 3x^2}{2}\ \mathrm{d}x = \frac{5}{8}.$$

Therefore, by Theorem 4.7,

$$\boxed{\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] = \frac{5}{4}}.$$

Also,

$$\boxed{\mathbb{E}[XY] = \int_0^1 \int_0^1 xy \frac{3}{2}(x^2 + y^2)\ \mathrm{d}x\ \mathrm{d}y = \frac{3}{8}}.$$

(c) Again by symmetry,

$$\mathbb{E}[X^2] = \mathbb{E}[Y^2] = \int_0^1 x^2 \frac{1 + 3x^2}{2}\ \mathrm{d}x = \frac{7}{15}.$$

Therefore,

$$\mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{7}{15} - \left(\frac{5}{8}\right)^2 = \frac{73}{960},$$

and, by Theorem 4.4,

$$\mathrm{Cov}(X, Y) = \frac{3}{8} - \left(\frac{5}{8}\right)^2 = -\frac{1}{64}.$$

Thus,

$$\boxed{\mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{73}{960}, \qquad \mathrm{Cov}(X, Y) = -\frac{1}{64}}.$$

(d) Using the moments already found,

$$\mathbb{E}[(X + Y)^2] = \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] = \frac{101}{60},$$
$$\mathrm{Var}(X + Y) = \frac{101}{60} - \left(\frac{5}{4}\right)^2 = \boxed{\frac{29}{240}}.$$

□